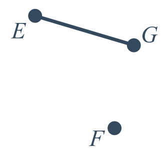
Line segments and distance

1



2 $\overline{AC}$

3 $\overline{XW}$

4 $\overline{AE}$

5 $\overline{DF}$

6

a 5 units

b 5 units

7

a 4 units

b 1 unit

c $c = \sqrt{17}$

8

a 2 units

b 1 unit

9 $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

10 Distance $= \sqrt{(-5-4)^2 + (10-(-5))^2}$

11 10.00

12

a 15 units

b $\sqrt{73}$ units

13 $\sqrt{34}$ units

14

a $\sqrt{82}$ units

b $\sqrt{26}$ units

c N

15

a $\sqrt{50}$ units

b 5 units

c M

16

a $\sqrt{50}$ units

b $\sqrt{2}$ units

c $5:1$

17

a $CE = 28$

b $GH = 15$

18

a $QS = 63$

b $PQ = 24$

19

a $x = 3$

b 25

20 $JT = 40$

21

a

i $x = 3$

ii 6

b

i $x = 3$

ii 3

Additional questions

22

a

i $AB = 3$ units

ii $BC = 4$ units

iii $AC = 5$ units

b

i $AB = 4$ units

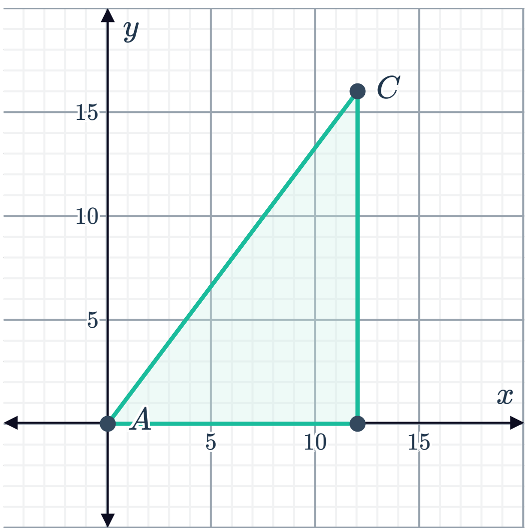
ii $BC = 7$ units

iii $AC = 8.06$ units

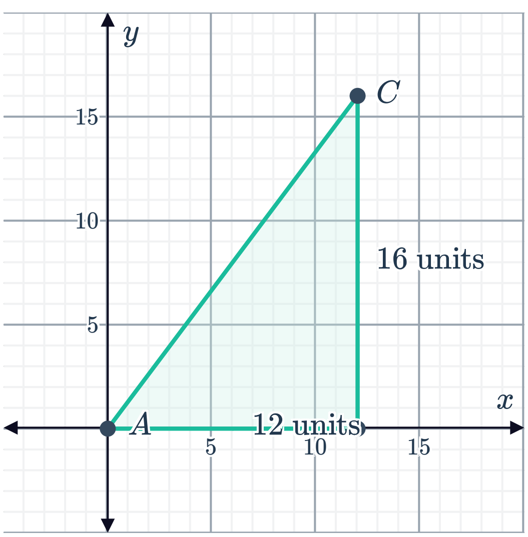
23

a

i



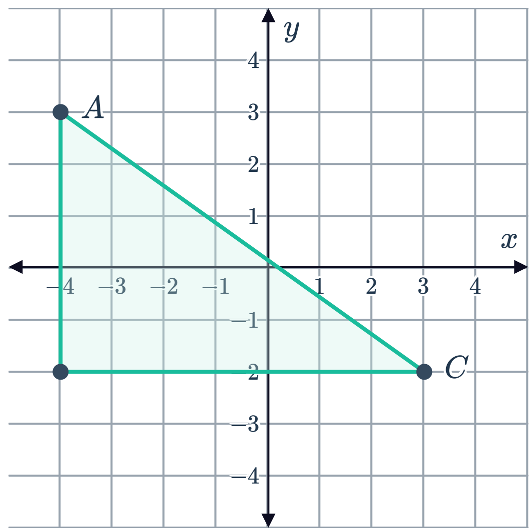
ii



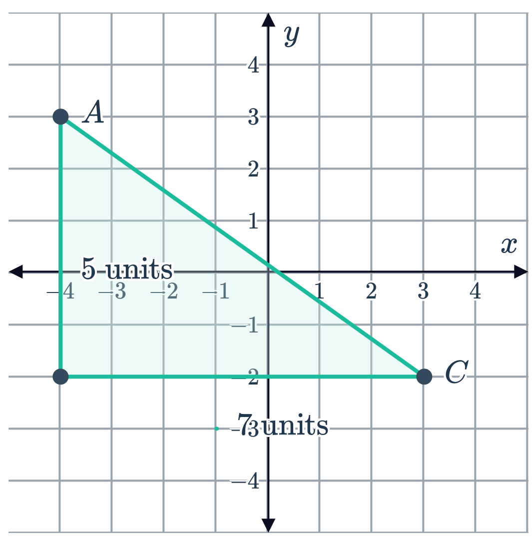
iii $AC = 20$ units

b

i



ii



iii $AC = 8.6$ units

24

25

a 10 units

b 15 units

c 8.54 units

d 5.83 units

26

a $PM = 9.90$ units

b $PN = 11.18$ units

c N

27 17 units

28

a $AC = 14.14$ units

b $BD = 8.49$ units

29

a Culmer Station is the closest to their meeting spot.

b When actually walking, we need to consider traffic lights, side walks and what is accessible. Depending on which direction they are catching the train may also impact their choice.

c If they need to stop at the Winn-Dixie, then the Civic Center Station would have the shortest trip distance.

30

a There are two possible coordinates.

b $M\,(-2, 9)$ and $M\,(-2, 3)$

c If we were not told the line segment was vertical, there would be an infinite number of answers.

31 The distance formula comes from the Pythagorean theorem. $(x_2 - x_1)$ is the calculationg for the length of the line segment which is the horizontal component of the right triangle we can create. $(y_2 - y_1)$ is the calculationg for the length of the line segment which is the vertical component of the right triangle we can create.

32 Both Dayanna and David are correct. $(x_2 - x_1)$ and $(x_1 - x_2)$ will have the same magnitude, but different signs. However, when we square them, they will both become positive, so they will give the same answer.

33

a. Fyfe made an error on Step 2. They incorrectly substituted $(x_2 - y_2)$ and $(y_1 - x_1)$, but this calculation does not connect to the Pythagorean theorem to make a right triangle and so does not find the distance.

b.

1	$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$	Distance formula
2	$= \sqrt{(3 - -4)^2 + (-5 - 7)^2}$	Substitution Property of Equality
3	$= \sqrt{(7)^2 + (-12)^2}$	Simplify
4	$= \sqrt{49 + 144}$	Simplify
5	$= \sqrt{193}$	Simplify

34

a. $AC = 1$

b. $AE = 4$

c. $EF = 1$

d. $FB = 2$

e. $DB = 4$

f. $AB = 7$

35

a. By the segment addition postulate we know that $CD + DE = CE$ and so $CE = 28$

b. By the segement addition postulate we know that $FG + GH = FH$ and so $GH = 15$

36

a. $PR = 5.1 \text{ mm}$

b. $AB = 51.4 \text{ cm}$

37

a. $\overline{DE}$ and $\overline{EF}$

b. $\overline{CF}$ and $\overline{DB}$

c. In part (a) you would find two new congruent segments, $\overline{CG}$ and $\overline{DG}$, and in part (b) you would not find any additional congruent segments.

38

a $KL = 25$

b $JT = 40$

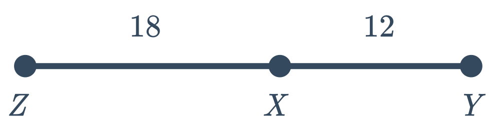
39

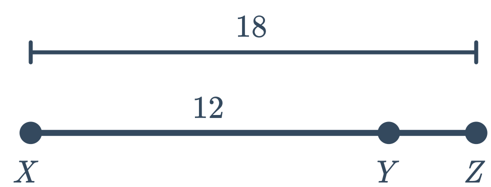
a $BD = 12$

b $CB = 3$

40 $ML = 51$

41 YZ could be 30 units in length or 6 units in length:





42 By the segment addition postulate, if points A, B, and C were collinear, then $AB + BC = AC$.
Since $AB + BC = 15$ and $AC = 12$, we know that these points cannot be collinear.